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Task Description:

The overall hypothesis driving the proposed research is that extended exposure to microgravity (μG) has a negative impact on the capacity of males of mammalian species to reproduce. This hypothesis is based on previous published work from our laboratory, demonstrating that hind limb sus-pension (HLS) of adult male rats for 6 weeks causes a block of spermatogenesis leading to permanent sterility. Furthermore, the initial 6 week HLS treatment triggers a decline in testicular function that continues beyond the initial HLS treatment period resulting in massive apoptosis of the germinal epithelium and sterility when assessed 6 mos later. One of the initial insults to the testis caused by HLS is a reduction in blood flow to levels just above the minimum known to sustain spermatogenesis. Essential components of this productive experimental approach have been to utilize i) partial inguinal to prevent the testes from becoming abdominal during hind limb elevation, and ii) pair-fed control animals identical to the HLS animals with inguinal ligation and tail harnessing, but elevated only to lift the tail but not limit contact of the hind limbs with the cage floor. New goals raised by the results during the previous project period are 1. Determine the maximum duration of HLS that permits subsequent restoration of spermatogenesis and fertility. 2. Ascertain the role that reactive oxygen species (ROS) play in the mechanisms that block spermatogenesis in long term HLS, and whether dietary administration of inhibitors of ROS damage serve as a countermeasure to the loss of spermatogenesis 3. Determine if prepubertal testicular development and adult testicular function are compromised in hypergravity (HG). Two of the Bioastronautics Roadmap (BR) risks are highlighted by our research:

• Risk 28: Increased cancer morbidity or mortality risk in astronauts may be caused by occupational radiation exposure – May be influenced by other space flight factors including microgravity & environmental contaminants. Mars mission not feasible unless improved shielding is developed. – Exposure to space radiation increases the risk of developing cancer later in life.

• Risk 30: Chronic and Degenerative Tissue Risks – Radiation exposure may result in degenerative tissue diseases. May be caused by occupational radiation exposure or the combined effects of radiation & other space flight factors – Acute or chronic illness due to tissue degeneration may lead to mission impacts, or adverse health consequences after return.

A key question raised by our results is whether the long term hindlimb unloaded model is predictive of the effect of long term microgravity on male reproductive health. This health concern is a separate but related issue to the increased risk of gonadal cancer due to radiation exposure during space flight. It should be noted that there is extensive literature demonstrating that the risk of gonadal cancer is increased by hyperthermia in the cryptorchid model. It is difficult to ascertain whether the time interval between testicular hyperthermia and the occurrence of testicular cancer would be within the time frame of a 30 month mission to Mars. In this regard, the onset of testicular cancers is postulated to be triggered in the post-cryptorchid male after puberty because of the influence of the hormonal milieu. Thus in the adult male with chronic testicular hyperthermia, this interval could be decreased because the adult male hormonal milieu is already present. When combined with increased radiation exposure, this will significantly elevate the risk for testicular cancer (at least 4-10 fold based on the cryptorchid model). In any case, because of the lack of data or investigations focused on the conditions experienced in space flight, these questions warrant more detailed investigation under controlled conditions.

This concern raised by our studies is relevant for human male astronauts in the context of the Bioastronautics Critical Roadmap as well as for rodents as an experimental model for humans. Unfortunately, no animal data is available because the 6 weeks hindlimb unloading is yet to be paralleled by space flight experiments of at least equivalent length. Human data from male astronauts or cosmonauts exposed to Microgravity for longer than Space Shuttle mission duration has not been published or been made available in response to gather such information. Thus it is impossible to ascertain whether or not similar problems have been reported in humans exposed to long term microgravity conditions.

A key question raised by these studies is whether the hindlimb unloaded model is relevant as a model for microgravity in males with scrotal testes. The regulation of testicular temperature well below body temperature is critical for the maintenance of spermatogenesis. The testicular temperature is regulated by a combination of blood flow and position relative to the body. Two opposing forces regulate the position of the testis: 1) the gravitational vector pulls the testicular mass away from the body promoting testicular cooling; 2) the cremasteric muscles pull the testes towards the body thereby increasing testicular temperature. This mode of temperature regulation is identical in all scrotal males including but not limited to humans, primates and rodents. Absence of the gravitational pull on the testis in microgravity eliminates the opposing affect of the cremasteric muscles causing the testes to remain elevated near the body; this is mimicked in the hindlimb unloaded animal because the head downward tilt position prevents gravity from pulling the testis away from the body. Thus microgravity and hindlimb unloading both prevent the testes from cooling properly, causing testicular hyperthermia and compromised spermatogenesis.

Thus, we hypothesize that in long term space flight, chronic testicular hyperthermia may occur in both rats and humans. In relation to the BR, this can cause significantly increased health risks not only for male infertility or sterility, but also for increased testicular cancer due to the well known combined effect of testicular hyperthermia and radiation exposure. Ultimately, this model must be verified/tested by long term rodent and human space flight experiments, developing appropriate countermeasures, and monitoring male astronaut health post flight.

We are thus faced with the following biomedical question: what level of post-flight astronaut cancer and/or infertility risk is acceptable? These risks will magnify with the lengthening duration of space flight and exposure to radiation outside of low earth orbit. The problem of gonadal cancer could even occur within the time frame of a long duration mission such as the 2 year Mars expedition, and post-flight following missions of 6 months or longer on ISS or more likely to the moon. Sterility and subfertility would be manifested post flight in such cases. Is there a willingness of NASA to address these questions by animal and human experimentation, and to address countermeasure development and assessment?

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Male infertility represents approximately half of the cases of infertility in couples. Inflammatory responses to hyperthermia and/or altered blood flow are a common component of male infertility of a broad spectrum of etiologies. The long term hindlimb unloading model appears to reproduce many of the same alterations in testis histopathology associated with many male infertility scenarios. Our studies in relation tp specific aim 3 in particular will study the efficicacy of potential countermeasures for inflammatory and post-iscemic induced male infertity regardless of whether in microgravity or in earth normal gravity.

Task Progress:	Increasing mechanical load to 2G in a 1G environment has a negative effect on bone quality of rats Bone loss due to reduced gravity is a major problem in spaceflight and increased mechanical loading created by artificial gravity is a potential countermeasure. In this study, 24 adult male Fischer 344 rats were randomly assigned to three groups: 1) Vivarium Control (VC) – Fed ad libitum ; 2) Stationary Control (SC) - Pair-Fed to 2G; and 3) 2G Group (2G). Mechanical load was set to 2G by housing animals on a 24-ft diameter centrifuge at NASA Ames Research Center. Final body weights after 45 days were 346, 325, and 287 g respectively indicating that centrifugation reduced body weight by mechanisms in addition to reduced food intake. Microarchitecture of the isolated distal femur was evaluated with a microCT 40 scanner; one hundred 16.5 micron slices of cancellous bone were contoured as a volume of interest (VOI). Finite element analysis predicted bone strength. Cortical thickness of the femoral midshaft and connectivity density and trabecular number (TbN) of the VOI were reduced and trabecular separation (TbSp) was increased in the SC compared to the VC. Strain, force to compress the VOI, and stiffness were decreased and von Mises stresses increased in SC compared to VC. Bone volume fraction and TbN decreased in 2G compared to SC and TbSp was increased. Strength parameters including force and stiffness tended to decrease ($p<0.07$) and von Mises stresses tended to increase ($p<0.08$) in 2G compared to SC. Increasing mechanical load to 2G by continuous centrifugation may be too great a force for optimal bone quality. (Supported by NASA and NIH). Male reproductive status in rats is unaltered under continuous long term artificial gravity (AG) The Bioastronautics Roadmap (BR) risks include tissue degeneration caused by occupational radiation exposure or the combined effects of radiation and other space flight factors. One potential countermeasure to the effects of microgravity (µG) is artificial gravity (AG) using centrifugal forces. Sexually mature male rats were maintained for 6 weeks at 2G (AG) in the NASA/ARC 24ft centrifuge facility. Stationary controls (SC) were pair-fed to the AG animals. A group of ad lib fed vivarium controls (VC) was maintained for the same period. At the end of 6 wks, half the animals were sacrificed to obtain testes for histology and blood for hormone analysis. The remaining animals were subjected to mating trials to determine whether fertility had been affected by the continuous AG treatment. Immediately at the completion of the AG exposure, testis weights were significantly greater (as % of whole body weight) in the AG group versus the SC and VIV groups. In the mating trials, all treatment groups were fertile. Similar to the increase in testis weight of the AG group, there was a significant increase (+1.4, $P<0.02$) in the number of pups per litter produced by the AG group versus SC. No other parameters of fertility were different between any of the treatment groups. By the completion of the mating trials (11 wks after terminating AG), the difference in testis weight had disappeared between the groups. The testis weight and litter findings are particularly interesting in that they are consistent with the hypothesis that biologic systems respond to gravitational forces along a continuum. In this case, increased gravity produced an increase in litter size and testis weight whereas we previously reported that simulated long term µG, using the hindlimb suspension model, produced sterility and a decline in testis weight. These results also provide preliminary data for AG as a potential countermeasure for male infertility problems associated with µG. However, the radiation exposure issue in regard to testicular cancer risk is another problem that will require additional environmental (shielding) and pharmacologic interventions. (Supported by NASA and NIH). Negative impacts of altered gravity models on male mammalian reproductive capacity Previous research showed that simulated microgravity using hindlimb unloading (HLS) for six wks leads to blockage of spermatogenesis and permanent sterility in adult male rats. Additionally, the exposure to HLS leads to a decline in testicular function causing massive apoptosis of the germinal epithelium. The goal of this project was to determine the maximum duration of HLS that allows for a subsequent restoration of spermatogenesis and fertility. Sexually mature male rats underwent partial inguinal ligation to prevent the testes from becoming abdominal during HLS. Control animals (TO) were pair fed to the HLS group and also ligated and harnessed, but the hindlimbs maintained contact with the floor. Additionally, a group of free roaming control animals (FRC) was analyzed. Animals were exposed to HLS, TO, and FRC treatment for 4, 5, and 6 wks. Hormone levels of testosterone, FSH, LH, and corticosterone were determined, and as in previous studies, none of the hormone levels were significantly different in the treatment groups. However, there was a significant spike in spermatogenic cell apoptosis at 6 wks, indicating that losses of spermatogenesis were beginning. In this experiment, no significant difference in testis or epididymal weight was noted with time (ANOVA). Since these animals were younger than in previous studies, we conclude that there may be an age-related sensitivity of the testis in the HLS model. (Supported by NASA and NIH). Hypergravity delays pre- and post-natal testicular development of male rats We are currently continuing analysis of data obtained from an experiment conducted at the NASA Ames Research 24 ft centrifuge facility. In that experiment, pregnant female rats were exposed to continuous 2G from gestation day 9 through birth of litters, and the resulting pups continued to be maintained under continuous 2G with either 2G-conditioned (HG) or non-conditioned foster mothers (HGN) until day 21 of post-natal development. Two days after birth, all litters were adjusted to contain 8 males and 2 females to ensure equivalent nursing and nutritional load on the mothers. Parallel control litters consisted of stationary controls (SC) and rotational controls (RC) that were maintained near the center of the centrifuge. Both body weight and testis weight were significantly lower in HG and HGN pups at 21 and 45 days of age when compared to the SC and RC litters. Mating trials at 45 days of age produced no litters in any of the treatment groups, whereas by day 60, all treatment groups produced F1 litters with no significant difference in the number of pups, gender distribution or pup weight measured at 5 days of age. The significantly lower testis size in the HG and HGN animals at 21 and 45 days of age are currently being processed for histologic analysis of spermatogenesis to determine if there was a delay in puberty when compared to the SC and RC control groups. In addition, gene array analysis of the testes are also being perfomed to examine for potential differences in gene regulation in the testes for the HG and HGN groups in comparison to the SC and RC controls.
Bibliography Type:	Description: (Last Updated: 01/24/2022)
Abstracts for Journals and Proceedings	Lessley J, Stoecker BJ, Globus RK, Baer L, Tash JS . "Increasing mechanical load to 2G in a 1G environment has a negative effect on bone quality of rats." Experimental Biology, San Francisco, CA, April 2006. Experimental Biology 2006 abstracts, April 2006. , Apr-2006
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